Information Management
SE 3: Information Systems Development

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Slides based on Prof.dr.ir. Hajo Reijers

Agenda

1. Quiz Questions
2. Make or Buy – Reasons for System Development
3. Systems Development
4. Systems Analysis Phase

Intermediate Exam Questions

Exemplary Exam Questions

Which of the following is among the key features of business intelligence tools?

a) Discarding unstructured data to achieve reliable analysis.

b) Disabling the availability of internal data to anyone outside the organization.

c) Utilization of both internal and external inputs.

d) Acting as a mainstream IT investment that does not help to create competitive advantage.

More on the Intermediate Exam

- Multiple-choice questions on the concepts of first three weeks.
- Accounts for 40% of overall grading.
- Your book for this part of the course is **"Business Information Management - Improving Performance Using Information Systems" by Chaffey and Wood** and **7th chapter of the book "Business Information systems" by Curtis & Cobham.**
- You are responsible for sections listed as "Essential (exam material)" in the weekly study instructions.

Make or Buy – Reasons for System Development

What is IS Development?

- Refers to the process of planning, creating, testing, and deploying an information system

Idea

Analysis

Design and Implementation

Completed Information System

Reasons for Project Initiation

- Current system cannot cope
- Cost savings
- Provision of better internal information for decision making
- Provision of better customer service
- Opportunities provided by new technology
- High-technology image
- Change in legislation
- Balancing the portfolio of projects

Make or Buy?

- **Build Software**
  - Customized
  - Developed in-house
  - Developed by outside vendor
  - Full control can be beneficial if software is strategically important

- **Buy Software**
  - Prepackaged software
  - Select software vendor
  - Customization may be offered by software vendor for a price

Making can be problematic

≡ Menu nrc.nl

Digitalisering rechtspraak is mislukt en moet helemaal opnieuw

Miljoenenproject Aan het digitaliseren van de rechtspraak is al 220 miljoen euro besteed en nu is het project mislukt. Het nieuwe, beperktere, doel: burgers en advocaten in een beveiligde omgeving met de rechtbank laten communiceren.

Mark Lievisse Adriaanse 10 april 2018

Digitization of the judiciary has failed and must be done all over again

KEI had onder meer tot doel digitaal procederen mogelijk te maken. Dat is voorlopig van de baan. De doelen van KEI waren te ambitieus en de uitvoering te ingewikkeld om het project te laten slagen, schrijft de Raad. Onderschat is hoe ingewikkeld het digitaliseren van de rechtspraak bleek te zijn, omdat het werk binnen de rechtspraak "te complex" is. Bovendien was

It is underestimated how complicated the digitization of the judiciary turned out to be, because the work within the judiciary is "too complex".

Rep

Het nieuwe doel is nu om het voor burgers en advocaten mogelijk te maken in een beveiligde omgeving met elkaar te laten communiceren. Het gaat volgens de Raad nu eerst om "toegankelijkheid". Een woordvoerder: "De administratie wordt digitaal, maar niet meer geautomatiseerd."

Al bij de start van het project is "niet voldaan" aan voorwaarden om het tot een succes te maken. Zo waren "de werkprocessen bij aanvang van het programma niet geharmoniseerd". Daar

inge

For example, "the work processes were not harmonized at the start of the program"

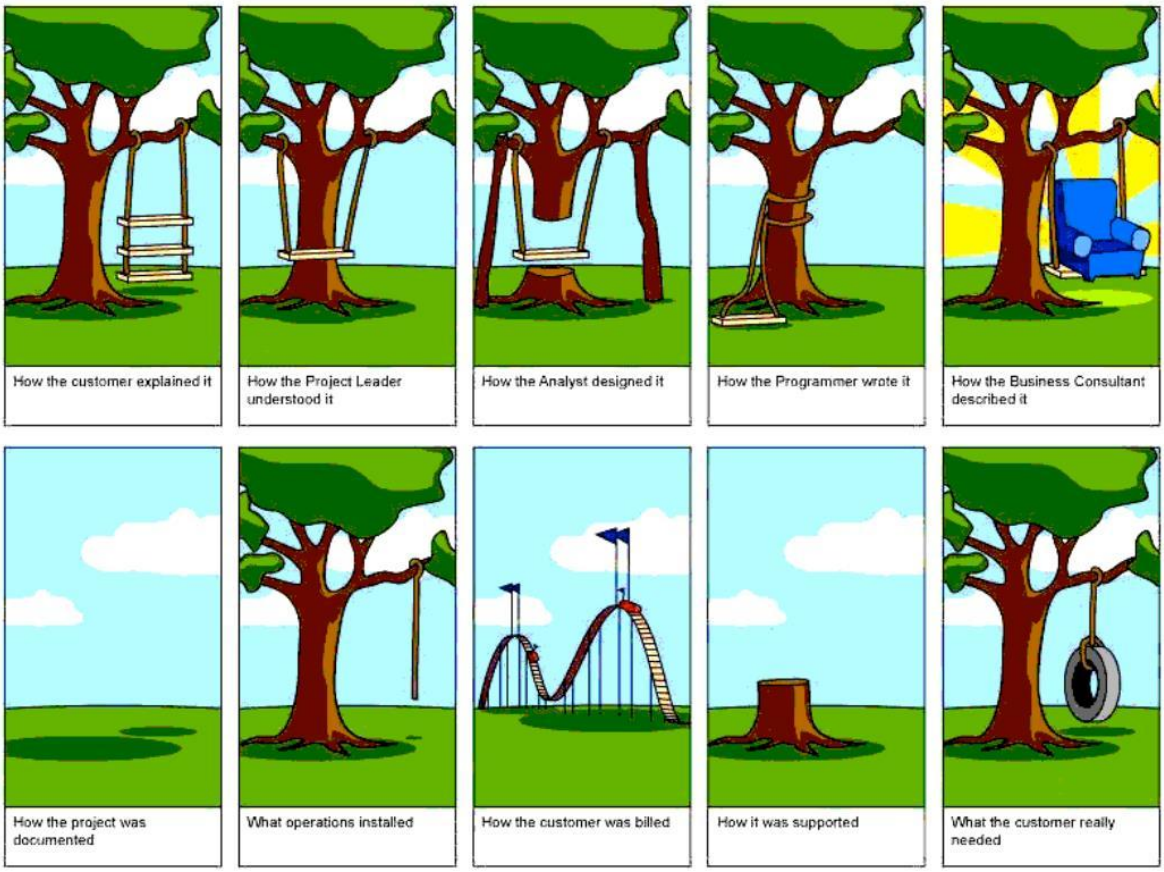
KEI werd daardoor ook veel duurder. In 2012 werden de kosten nog rond 7 miljoen euro geschat, inmiddels is al 220 miljoen euro uitgegeven. Volgens een brandbrief van de Centrale Onderneming

proj

In 2012, the costs were estimated at around 7 million euros, while 220 million euros have already been spent.

In de nieuwe aanpak worden kortere projecten opgezet met minder ambitieuze doelen voor een beperkter deel van de organisatie. "Digitalisering is onontkoombaar", schrijft de Raad. Hoeveel geld het nieuwe project gaat kosten, wordt volgens een woordvoerder in de loop van 2018 duidelijk.

Making can be problematic



Buying can be problematic

Artikelen | Community | Carrière | Topics | Awards | NEXT | Whitepapers | Meer | Service

Home | Artikelen | Nieuws | ERP

Lidl neemt verlies van half miljard op SAP-project

'eLWIS is dood'

Lidl takes half billion loss on SAP project

26 september 2018 16:31 | Cees Visser | 4

Topic ERP

Lidl is circa een half miljard euro kwijt aan een mislukt SAP-project. De Duitse supermarktketen is sinds zeven jaar bezig met de implementatie van de bedrijfsapplicatie, maar heeft het vertrouwen in een goede afloop verloren. Dat schrijven Duitse media.

Om de it-infrastructuur te vereenvoudigen, startte Lidl in 2011 met de implementatie van het zogenoemde eLWIS-systeem (naar het Duitse Elvis), gebaseerd op SAP HANA for retail. Dat verliep aanvankelijk goed: het systeem werd zelfs inter-

Costs to consider when making

- Initial cost of building a system
- Cost of enhancing the system with new features
- Cost of repairing defects and bugs
- Cost of customer support
- Cost of refactoring
- Removal of error-prone modules

Source: Jones, C. (2007). Geriatric Issues of Aging Software. *Crosstalk*, Dec. 2007.

Costs to consider when buying

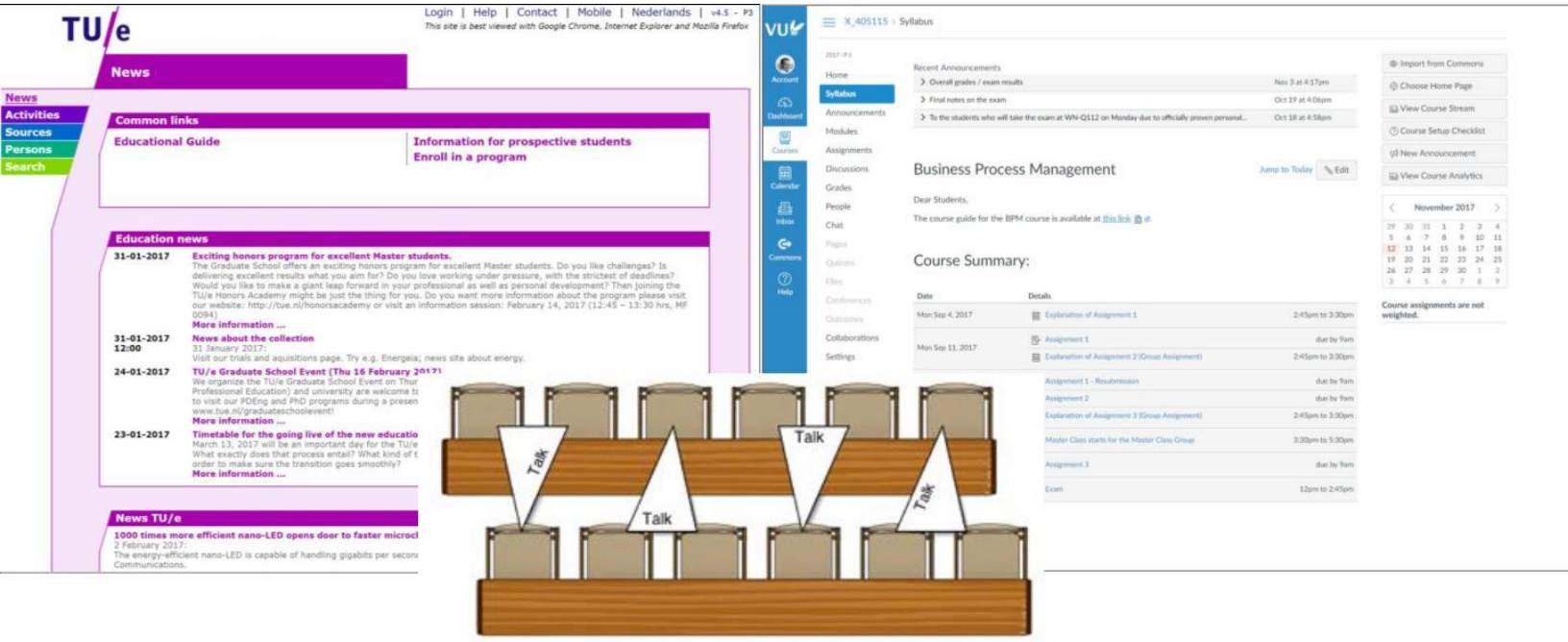
- Cost to purchase software
- Cost to customize software
- Licenses for named or concurrent users
- Cost of infrastructure that software will run on: servers, network connections, network upgrades to support higher loads or number of transactions
- Staffing: training, maintenance, support, administration, incidental personnel

When does “Making” make sense?

- Making software may be associated with risks
- Carefully considering potential costs is essential
- Strategic software is custom software
- In-house development is even more critical if you wish to retain your firm’s advantage

VU Canvas vs TU/e OASE

- Group Exercise:
- What are the factors for making and buying for Learning Management Systems (LMS)?



Factors for Making or Buying LMS

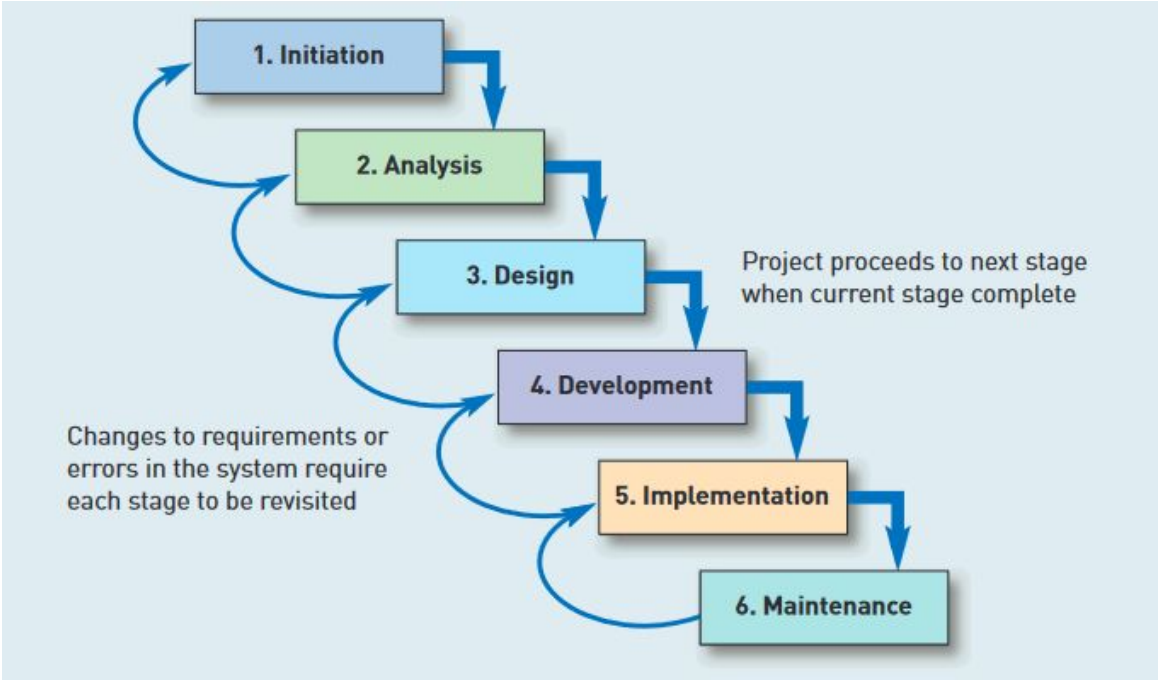
Making	Buying
Advantages	
Can be customized	Product can be tried
Full-control and strategic advantage	Possible to choose among vendors
No dependency on external vendors	Customization can be possible
Can be interconnected to other systems	Possible to extend/connect to existing systems
Risks / Concerns	
High risks for defining requirements and the cost	Cost is clear, but may require high initial investment
The need for full-support and maintenance	Responsibilities can be transferred to the vendor
The need to keep up-to-date with the technology	License limitations

Systems Development

Systems Development Process

- **Initiation** → Start up phase
- **Analysis** → Define Requirements
- **Design** → Specify System Configuration
- **Development** → Programming and configuring the system
- **Implementation** → Installing and testing the system
- **Maintenance** → Monitoring and revising the system

Systems Development Process : The waterfall model



Systems Development Process : Prototyping (1/3)

Prototyping is a common approach to developing information systems which is used to counter the problems of the waterfall model. The essence of prototyping is that it is:

- Rapid
- Simple
- Iterative
- Incremental
- User-centred

Systems Development Process : Prototyping (2/3)

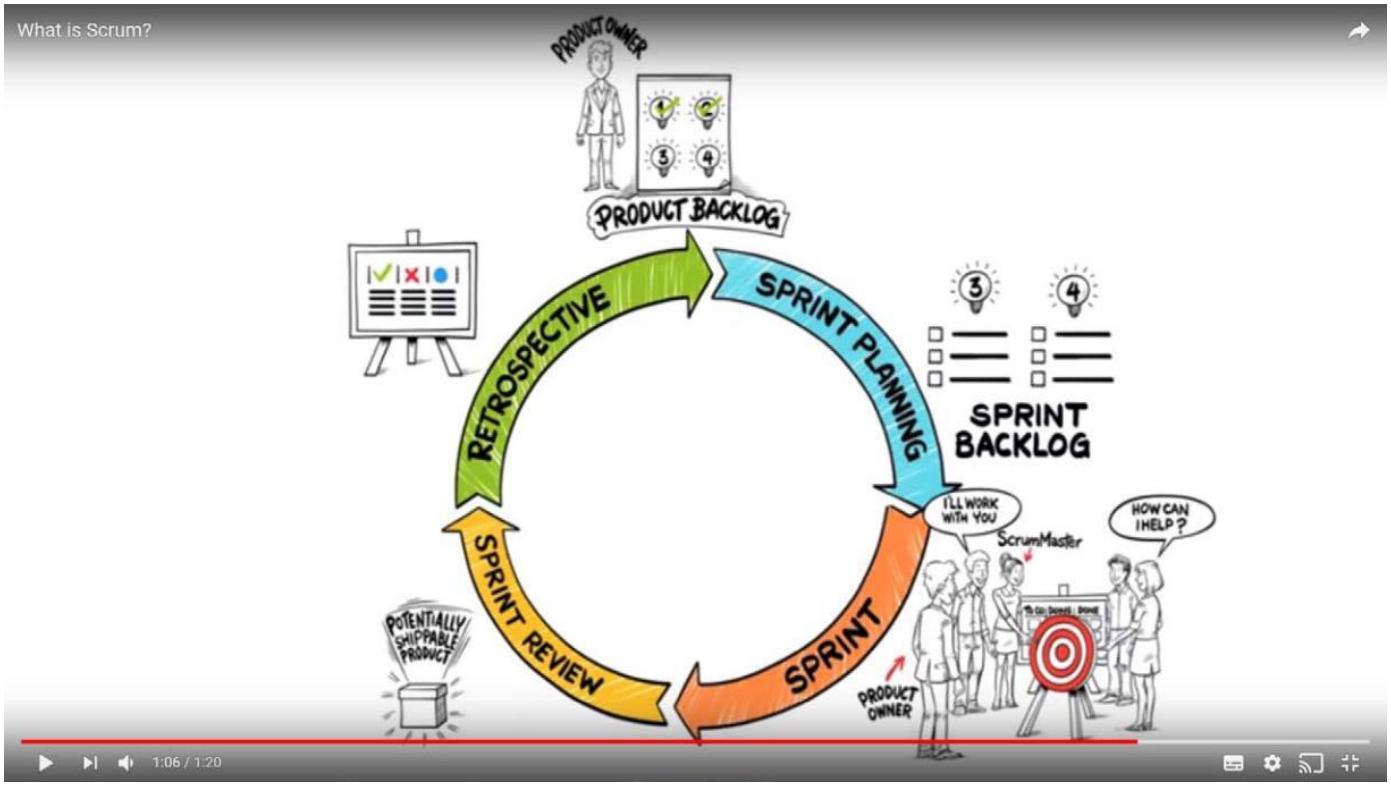
- Once the first prototype has been produced there are several alternatives:
  - Iterate and produce further refinements, which often occurs throughout the specification stage
  - Develop module prototypes
  - Throw away the prototype and develop a more robust version of the software for the production version

Systems Development Process : Prototyping (3/3)

Repeated iterations of the analysis, design, develop, test and review stages occur during prototyping. So, the approach to prototyping should be considered carefully. Preece et al. (2002) identify three different types of prototyping:

- Throwaway or low-fidelity prototyping
- Evolutionary prototyping or high-fidelity prototyping
- Incremental prototyping

Alternatives: Agile -> Scrum

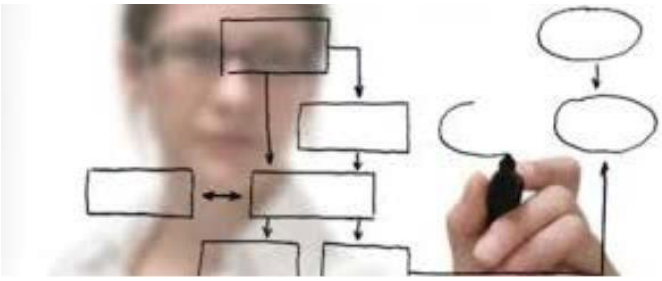


https://www.youtube.com/watch?v=TRcReyRYIMg

Systems Analysis Phase

Systems analysis

- Goal is to build a logical model of the system
- Question: What must be done in order to satisfy the objectives and functions of the system under investigation?
- Two main parts of analysis:
  - *Data analysis*: Which data has to be stored
  - *Process analysis*: Aims at showing the activity flow of the system, e.g. interactions with the system



Participants in Analysis and Design

Business Side:

User / Domain Expert / Client

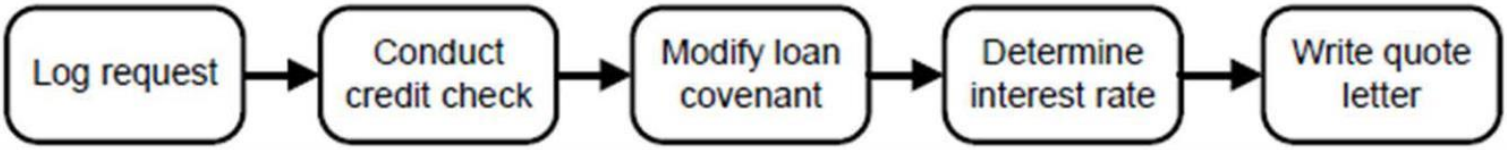
Mediator:

Business Analyst

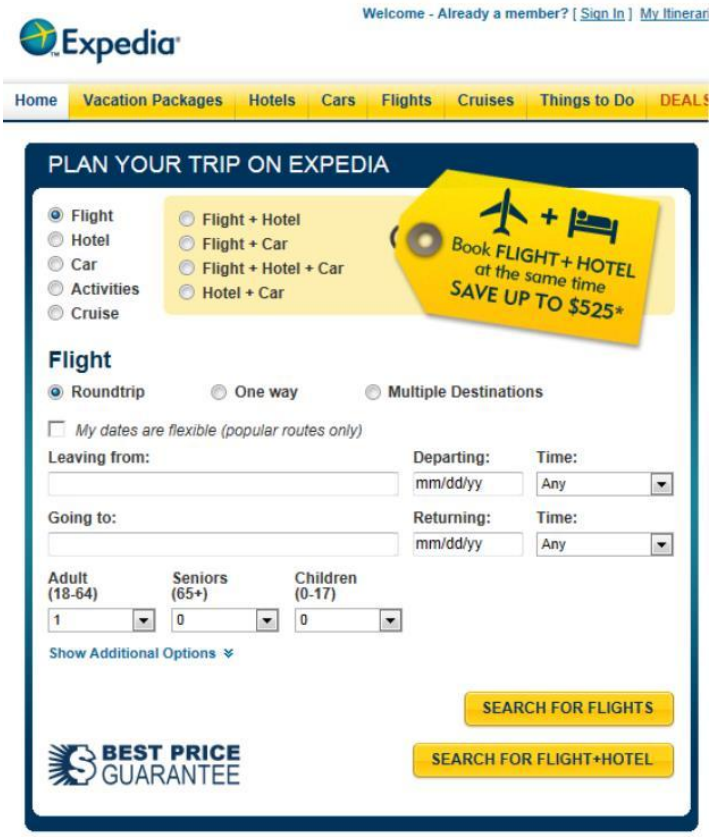
IT Side:

Developer / Programmer

Challenge 1:
Fragmented Process Knowledge



Challenge 2:
Domain Experts think on Instance Level



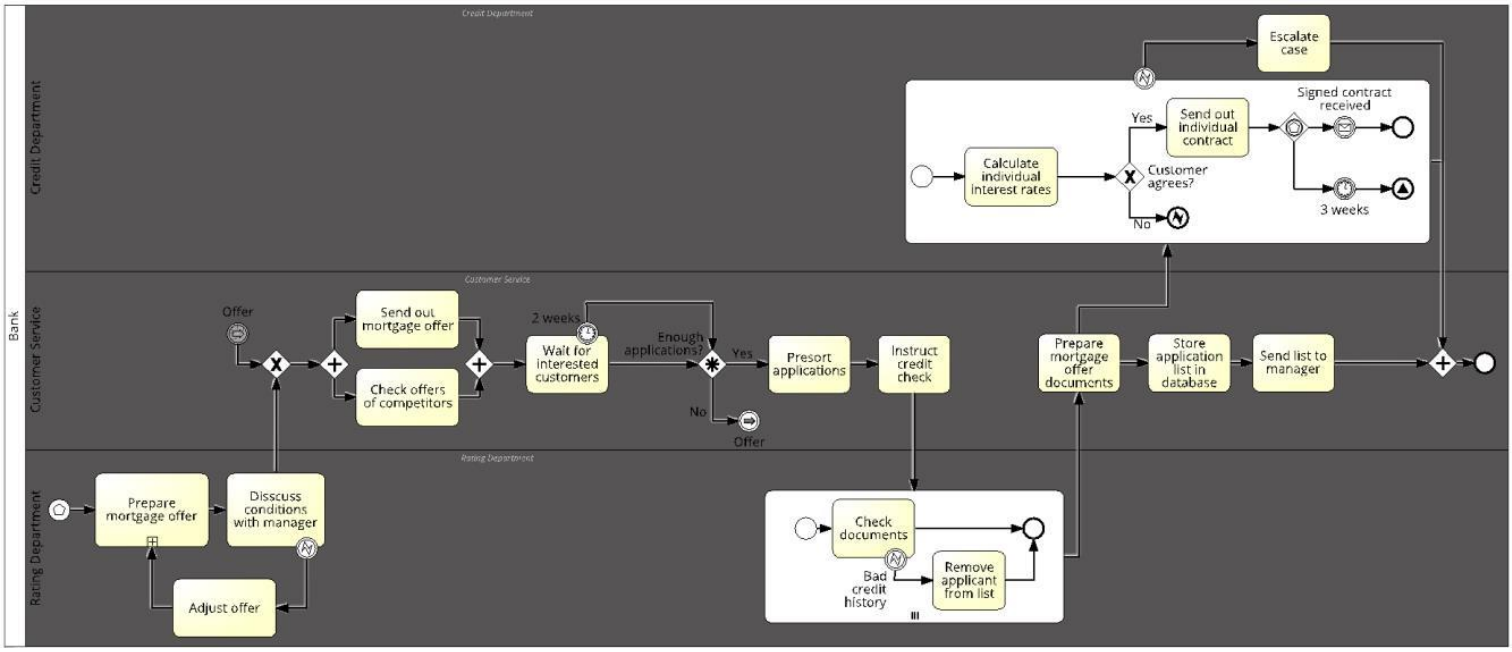
"Every trip is different."

"You cannot really compare. Our customers go to different places in different seasons using different modes of transportation."

"We can never do anything exactly in the same way. There are so many special conditions."

Challenge 3: Knowledge about Process Mining is rare

"Could you please tell me, whether this diagram correctly shows your process?"



Key aspects of System analysis

1. Focus Groups
2. Documentation
3. Surveys
4. Observation
5. Interviews
6. Prototyping
7. Producing the requirements specification

Focus groups

- Focus groups or brainstorming are useful at an initial stage.
- Helps to identify major organizational and process.
- Helps is identifying the issues which the project should address, and these can then be followed up in more detail with the other requirements-gathering techniques.

Documetation

A range of documentation may be available from existing systems. This may include:

- previous requirements specifications
- user guides
- procedure manuals or
- help systems.

Pre-existing good-quality documentation may make documenting the requirements of the new

Survey

- Questionnaires are sent to potential users of the system and users of the existing system.

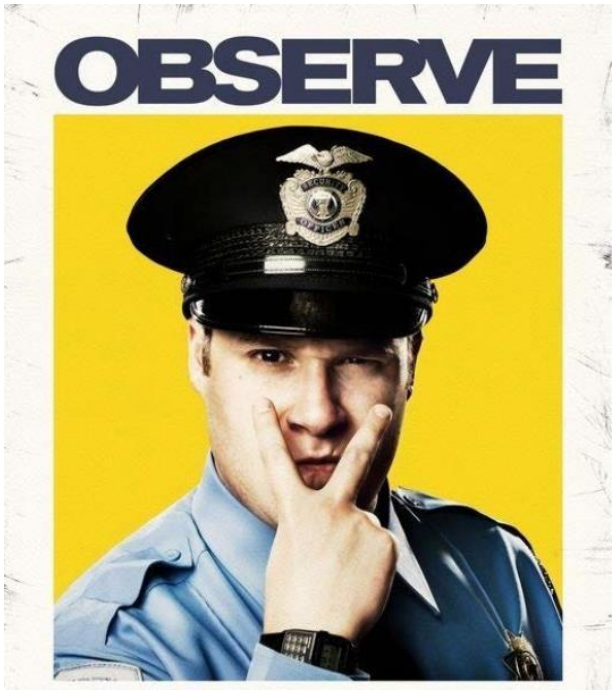
- These also provide a structured method of requirements gathering.

- It is possible to determine key features by asking respondents closed questions such as selecting from a list or asking more general open questions such as 'What are the three best and worst features of the current system?'.

- Surveys work best if performed before interviews

Observation

- Assessing how users of the current system work when completing different activities.
- It is an essential technique for identifying inefficiencies in a system.
- It is less clear at which point observation should occur in the sequence of requirements determination activities.
- It will often occur in parallel with other activities.

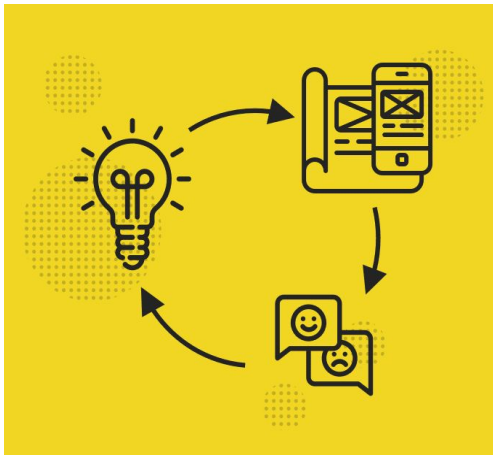


Interviews

- Interviews are a more structured form of collecting requirements than focus groups.
- They give an individual time to reflect on their needs and describe problems with the existing system.
- Detailed requirements are collected so it makes sense if these happen towards the end of requirements gathering.
- Some interviews with managers may be useful at an earlier stage in requirements determination.
- Interviews work best if they are semi-structured, with a prepared script of questions which must be answered but flexibility for other concerns to be described.

Prototyping

Prototyping of new systems through paper or on-screen prototypes will in itself elicit requirements.



Producing the requirements specification

The requirements specification will detail how the system should support its users and business processes. It will include:

- Business requirements – what are the required business outcomes of developing the new system?
- Summary of system users and their general requirements.
- Process support requirements – which work activities will the system support?
- Data output requirements – what information will be available to users and managers of the system?
- Data input requirements – what information will be captured for the system, what will its source be (other systems and manual data entry)?
- Data visualization requirements – how will data be entered and viewed by users?

To Remember for the Exam

To Remember for the Exam

- Make or buy: making can be risky but should be considered for strategically relevant systems
- We looked at different system development processes
- We looked at the different stages of system development process
- We looked in depth at the key elements of the systems analysis phase
- Homework : Read up on the other phases from the textbook